- G-set X.
  - $e \cdot x = x$
  - $(g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x)$

$G_x = \{ g \in G : g \cdot x = x \}$.

- $C_G(x) = \{ g \in G : gx = xg \}$. $\quad N_G(H) = \{ g \in G : gH = Hg \}$.

$\mathrm{orb}(x) = \{ g \cdot x : g \in G \} \subseteq X$.

- $$|\mathrm{orb}(x)| = \frac{|G|}{|G_x|} \iff |\mathrm{orb}(x)|\,|G_x| = |G|.$$

<u>Thm 13.3</u>. Let G be a finite group of order p^n, where p is prime. Then <u>$Z(G)$ contains more than one element</u>. ($Z(G)$ contains at least p elements.)

<u>Pf</u>. Let $X = G$ with $g \cdot x = gxg^{-1}$.

$$G = \mathrm{orb}(g_1) \cup \cdots \cup \mathrm{orb}(g_k)$$

$$\Rightarrow \quad |G| = |\mathrm{orb}(g_1)| + \cdots + |\mathrm{orb}(g_k)|.$$

$g_k = e \Rightarrow |\mathrm{orb}(e)| = 1$.

$$\Rightarrow \quad \underbrace{|G|}_{p^n} = \underbrace{|orb(g_1)|}_{p^{m}} + \cdots + \underbrace{|orb(g_{k-1})|}_{p^{m}} + 1$$

orbit-stabliser

$$\Rightarrow \quad p^n = p^{m_1} + p^{m_2} + \cdots + p^{m_{k-1}} + 1.$$

$\Rightarrow \quad \exists\, h \in G, h \neq e$ s.t. $orb(h) = \{h\}$. $\Rightarrow \quad ghg^{-1} = h \Rightarrow h \in Z(G)$.

<u>Thm 13.4</u>. Let G be a group such that $G/Z(G)$ is a cyclic group. Then G is abliean.

<u>Pf</u>. $G/Z(G)$ is a cyclic group, then there is $g \in G$, s.t.

$$G/Z(G) = \langle gZ(G) \rangle.$$

$$\boxed{(gZ(G))^n = g^n Z(G)}$$

$\forall\, g_1, g_2 \in G$, $\exists\, n_1, n_2 \in \mathbb{Z}$, s.t. $g_1 = g^{n_1} z_1$, $g_2 = g^{n_2} z_2$, $z_1, z_2 \in Z(G)$.

$$g_1 g_2 = g^{n_1} z_1 g^{n_2} z_2 \overset{z_1 \in Z(G)}{=} g^{n_1} g^{n_2} z_1 z_2 = g^{n_2} g^{n_1} z_2 z_1 = g^{n_2} z_2 g^{n_1} z_1 = g_2 g_1.$$

$\Rightarrow$ G is abelian.

<u>Thm 13.5</u>. Any finite group G with $|G| = p^2$ elements, p is prime, $\boxed{\text{is abelian}}$.

<u>Pf</u>. $Z(G) \trianglelefteq G \Rightarrow |Z(G)| = 1, p, p^2$

Thm 13.3 $\Rightarrow$ $|Z(G)| > 1$.

$|Z(G)| = p^2 \Rightarrow Z(G) = G \Rightarrow G$ is abelian.

$|Z(G)| = p \Rightarrow |G/Z(G)| = p \Rightarrow G/Z(G)$ is cyclic.

Thm 13.6. $\Rightarrow$ G is abelian.

Lem 13.1. Let G and H be two subgroups of a finite group J. Then

$$|GH| = \frac{|G||H|}{|G \cap H|}$$

Pf. Note that $G \cap H \leq G$, consider left cosets

$g_1(G\cap H), g_2(G\cap H), \dots, g_n(G\cap H)$, $g_i^{-1}g_j \notin G\cap H$ $\forall i \neq j$ $\wedge$ $g_1 \cdots g_n \in G$.

$\forall\, gh \in GH$, $g \in G$, $h \in H$, $\exists\, g_i$, s.t. $g \in g_i(G\cap H)$.

$$gh = g_i \underbrace{(g^{-1}h)}_{G\cap H} = g_i(g^{-1}h) \in g_i H$$

$\Rightarrow$ $GH = g_1H \cup g_2H \cup \dots \cup g_nH$.

$g_iH = g_jH \Leftrightarrow g_i^{-1}g_j \in H \Rightarrow g_i^{-1}g_j \in G \cap H$. contradiction.

$\Rightarrow \quad g_i H \neq g_j H, \ \forall\, i \neq j.$

$$\frac{|GH|}{|H|} = n = \frac{|G|}{|G \cap H|}. \qquad \boxed{\checkmark}.$$

<u>Thm 13.6</u>. A group of order p^2 is isomorphic to either $\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.

<u>Pf</u>. $|G| = p^2$. If $\exists\, g \in G$, $\text{order}(g) = p^2 \Rightarrow G \cong \mathbb{Z}_{p^2}$.

If $\text{order}(g) = p$, $\forall\, g \in G$, $g \neq e$.

let $g, h \in G$ be such elements such that $\langle g \rangle \cap \langle h \rangle = \{e\}$

$\langle g \rangle \langle h \rangle = \langle h \rangle \langle g \rangle$

$\langle g \rangle \langle h \rangle$ is a group.

$$|\langle g \rangle \langle h \rangle| = \frac{|\langle g \rangle|\,|\langle h \rangle|}{|\langle g \rangle \cap \langle h \rangle|} = |\langle g \rangle|\,|\langle h \rangle| = p^2.$$

$\Rightarrow \quad \langle g \rangle \langle h \rangle = G \cong \mathbb{Z}_p \times \mathbb{Z}_p.$

$\Rightarrow \quad G \cong \mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.

$|G| = 4, 9, 25, 49, \dots$

$\mathbb{Z}_4$ or $\underline{V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2}$

$\mathbb{Z}_9$ or $\underline{\mathbb{Z}_3 \times \mathbb{Z}_3}$.

- $|G| \leq 7$.

$\underline{1}$	$\{e\}$
$\underline{2}$	$\mathbb{Z}_2$.
$\underline{3}$	$\mathbb{Z}_3$
$\underline{4}$	$\mathbb{Z}_4$, V_4.
$\underline{5}$	$\mathbb{Z}_5$
$\underline{6}$	$\mathbb{Z}_6$, $S_3 \cong D_3$.
$\underline{7}$	$\mathbb{Z}_7$.
• $\boxed{8}$	$\mathbb{Z}_8$, D_4 5
$\underline{9}$	$\mathbb{Z}_9$, $\mathbb{Z}_3 \times \mathbb{Z}_3$.

- $|G| = p$, p is prime, $G \cong \mathbb{Z}_p$.
- $|G| = \underline{p^2}$, p is prime, $G \cong \mathbb{Z}_{p^2}$, $\mathbb{Z}_p \times \mathbb{Z}_p$.

§14. The Sylow Theorems.

$|S_4| = 24$. $\quad |A_4| = 12$. $\quad 2 \times 6$

$\sigma = \underbrace{(i_1 i_2)(j_1 j_2) \cdots (l_1 l_2)}_{\text{even}} \leftarrow$ even

$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} = (23)(14)$ ~~odd~~ even.

$\sigma = (1234) = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$

$= (14)(13)(12)$. odd.

Exercise.

- A_4 does not have a subgroup of order 6.

$H \leq A_4$, $\quad |H| = 6$. $\quad H \cong \mathbb{Z}_6$ or S_3

A_4 $\quad 1$ $\quad$ ord$(1) = 1$. $\quad 3$ order 2 $\quad 8$ order 3

Def 14.1 Let p be a positive, prime integer. A p-group is a group in which every element has order a power of p. $\boxed{|G| = p^k}$

Exam 14.1 Any cyclic group of prime order is a p-group.

Exam 16.2. $|S_3| = 2\times 3$. $\langle a\rangle = \{e, a, a^2\}$ 3-subgroup.

$\{e, b\}$, $\{e, ab\}$, $\{e, a^2b\}$ 2-subgroup.

Def 16.2. Let G be a finite group with $|G| = mp^k$, $p\nmid m$. A subgroup of p^k is called Sylow p-subgroup.

Exam 16.3. $|G| = 60 = 2^2\cdot 3\cdot 5$

Sylow 2-subgroup (order 4).

Sylow 3-subgroup (order 3)

Sylow 5-subgroup (order 5)

Thm 16.1 (The Sylow theorem). Let G be a group of order mp^k $p\nmid m$, p is prime. Then

I. a Sylow p-subgroup (order p^k) exists.

II. for each prime p, the Sylow p-subgroups are conjugate to each other.

III. let n_p be the number of Sylow p-subgroups then

(i) $n_p \equiv 1 \pmod p$.

(ii) $n_p = \dfrac{|G|}{|N_G(P)|}$. $P \le G$ Sylow p-subgroup.

(iii) $n_p \mid m$.